

Introduction to Mathematics and Optimization

January, 2023

The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (7.11.2022) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = 3x + 2y$$

and $C \subseteq \mathbb{R}^2$ the subset given by

$$C = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 10, 2x + y \leq 15, x \geq 0, y \geq 0\}.$$

We consider the two optimization problems

$$\begin{aligned} \max & f(x) \\ & x \in C \end{aligned} \tag{1}$$

and

$$\begin{aligned} \min & -f(x) \\ & x \in C \end{aligned} \tag{2}$$

(a) Explain what the relationship is between the optimization problems (1) and (2).

ANSWER:

Since $f(x_0) \geq f(x)$ for all $x \in C$ if and only if $-f(x_0) \leq -f(x)$ for all $x \in C$, we see that $x_0 \in C$ is a solution to (1) if and only if it is a solution to (2). See also Remark 4.6.

(b) Sketch the set C and prove without reference to the sketch that C is a convex subset of $\mathbb{R}^2$.

ANSWER:

A sketch of C is produced by drawing the lines $x + y = 10$, $2x + y = 15$, $x = 0$, $y = 0$ and shading the intersection that arises from the inequalities:

EksIM023.png

The subset C is the intersection of the subsets

$$\begin{aligned} &\{(x, y) \mid x + y \leq 10\} \\ &\{(x, y) \mid 2x + y \leq 15\} \\ &\{(x, y) \mid x \geq 0\} \\ &\{(x, y) \mid y \geq 0\} \end{aligned}$$

each of which is a convex subset, since they all have the form $\{(x, y) \in \mathbb{R}^2 \mid f(x, y) \leq a\}$, where $a \in \mathbb{R}$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a convex function (Lemma 4.26). For example, the first subset can be expressed via $f(x, y) = x + y$ and $a = 10$. Here f is a convex function because its Hessian matrix is the zero matrix, which is positive semidefinite (Theorem 8.21). Hence C is an intersection of convex subsets and therefore convex (Exercise 4.22).

(c) Show that C is a compact subset of $\mathbb{R}^2$.

ANSWER:

The subset C is closed by almost the same argument as in part (b). It is an intersection of subsets of the form $g^{-1}([-\infty, a])$. Each of these is closed by Proposition 5.72 and Proposition 5.51. Hence C is an intersection of closed subsets and therefore closed (Proposition 5.52). Furthermore, C is bounded, since $x \geq 0, y \geq 0, x + y \leq 10$ implies that $|x| \leq 10$ and $|y| \leq 10$ and hence $x^2 + y^2 \leq 100$ for $(x, y) \in C$ (cf. §5.4.2). It is therefore compact.

(d) Which result in the curriculum ensures that the optimization problems (1) and (2) can be solved, and why?

ANSWER:

The function $f(x, y) = 3x + 2y$ (and $-f$) is continuous by Lemma 5.62 and Proposition 5.64. That the optimization problems can be solved is concluded from Theorem 5.76 and the previous part.

(e) Solve the optimization problem (1) using Fourier-Motzkin elimination.

ANSWER:

We follow the procedure outlined in §4.5. We set $z = 3x + 2y$ and first eliminate y via $y = \frac{1}{2}z - \frac{3}{2}x$. This is substituted into the inequalities that define C , and we obtain the inequalities

$$\begin{aligned} -\frac{1}{2}x + \frac{1}{2}z &\leq 10 \\ \frac{1}{2}x + \frac{1}{2}z &\leq 15 \\ x &\geq 0 \\ \frac{1}{2}z - \frac{3}{2}x &\geq 0 \end{aligned}$$

or

$$\begin{aligned}z - 20 &\leq x \\x &\leq 30 - z \\x &\geq 0 \\z &\geq x\end{aligned}$$

Now x can be eliminated and we end up with the inequalities $z - 20 \leq 30 - z$, $0 \leq z$, $0 \leq 30 - z$, $z - 20 \leq z$, i.e., $z \geq 0$, $z \leq 25$. The maximum value of z is thus 25. By substituting $z = 25$, we get $5 \leq x$ and $x \leq 5$, i.e., $x = 5$. Hence $y = (25 - 15)/2 = 5$. That is, the optimal value for the optimization problem is 25. This is uniquely attained at $(5, 5)$.

(f) Show that the KKT conditions corresponding to (2) have a solution.

ANSWER:

The KKT conditions have a solution by Theorem 9.30(i). The conditions for using this result are satisfied because the optimization problem has a solution and is strictly feasible (with e.g. $(1, 1)$ in the interior of C).

PROBLEM 2

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y) = x^2 - 4xy + 4y^2 - 6x - 3y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 10, 2x + y \leq 15, x \geq 0, y \geq 0\}.$$

We consider the optimization problem

$$\begin{aligned} \min f(x) \\ x \in C. \end{aligned} \tag{3}$$

(a) Show that

$$f(x, y) = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -6 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

ANSWER:

Here we use matrix multiplication and obtain

$$\begin{aligned} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -6 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \\ \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} x - 2y \\ -2x + 4y \end{pmatrix} - 6x - 3y = \\ x(x - 2y) + y(-2x + 4y) - 6x - 3y = x^2 + 4y^2 - 4xy - 6x - 3y = f(x, y). \end{aligned}$$

(b) Show that f is a convex function.

ANSWER:

The Hessian matrix (Definition 8.2) of f is

$$H = \begin{pmatrix} 2 & -4 \\ -4 & 8 \end{pmatrix}.$$

We use Theorem 8.21 and must show that H is positive semidefinite. This follows from Proposition 3.41 and Example 8.28: The matrix can be transformed with one operation to

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix},$$

which is positive definite according to Exercise 8.26.

(c) Write down the KKT conditions for (3).

ANSWER:

The KKT conditions for (3) are written as:

$$\begin{aligned}
 \lambda_1, \lambda_2, \lambda_3, \lambda_4 &\geq 0 \\
 x + y - 10 &\leq 0 \\
 2x + y - 15 &\leq 0 \\
 -x &\leq 0 \\
 -y &\leq 0 \\
 \lambda_1(x + y - 10) &= 0 \\
 \lambda_2(2x + y - 15) &= 0 \\
 \lambda_3(-x) &= 0 \\
 \lambda_4(-y) &= 0 \\
 2x - 4y - 6 + \lambda_1 + 2\lambda_2 - \lambda_3 &= 0 \\
 -4x + 8y - 3 + \lambda_1 + \lambda_2 - \lambda_4 &= 0.
 \end{aligned}$$

(d) Show using the KKT conditions that $(5, 4)$ cannot be an optimal solution to (3).

ANSWER:

Substituting $x = 5$ and $y = 4$ gives

$$\begin{aligned}
 \lambda_1, \lambda_2, \lambda_3, \lambda_4 &\geq 0 \\
 9 - 10 &\leq 0 \\
 14 - 15 &\leq 0 \\
 -5 &\leq 0 \\
 -4 &\leq 0 \\
 \lambda_1(9 - 10) &= 0 \\
 \lambda_2(14 - 15) &= 0 \\
 \lambda_3(-5) &= 0 \\
 \lambda_4(-4) &= 0 \\
 -12 + \lambda_1 + 2\lambda_2 - \lambda_3 &= 0 \\
 9 + \lambda_1 + \lambda_2 - \lambda_4 &= 0.
 \end{aligned}$$

The first four equations give $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$, which contradicts the fifth equation, which becomes $-12 = 0$. With reference to Theorem 9.31(i), we therefore cannot have an optimal solution for $x = 5, y = 4$. Note that the optimization problem is strictly feasible (in fact with $(x, y) = (5, 4)$ as an interior point of C).

(e) Use the KKT conditions to find a solution to (3). It may be assumed without proof that any optimal solution (x, y) to (3) has $x > 0$ and $y > 0$.

ANSWER:

The KKT conditions are

$$\begin{aligned}
 \lambda_1, \lambda_2, \lambda_3, \lambda_4 &\geq 0 \\
 x + y - 10 &\leq 0 \\
 2x + y - 15 &\leq 0 \\
 -x &\leq 0 \\
 -y &\leq 0 \\
 \lambda_1(x + y - 10) &= 0 \\
 \lambda_2(2x + y - 15) &= 0 \\
 \lambda_3(-x) &= 0 \\
 \lambda_4(-y) &= 0 \\
 2x - 4y - 6 + \lambda_1 + 2\lambda_2 - \lambda_3 &= 0 \\
 -4x + 8y - 3 + \lambda_1 + \lambda_2 - \lambda_4 &= 0.
 \end{aligned}$$

Here we see that $\lambda_3 = \lambda_4 = 0$ (since we have assumed $x > 0$ and $y > 0$) and the KKT conditions reduce to

$$\begin{aligned}\lambda_1, \lambda_2 &\geq 0 \\ x + y - 10 &\leq 0 \\ 2x + y - 15 &\leq 0 \\ -x &\leq 0 \\ -y &\leq 0 \\ \lambda_1(x + y - 10) &= 0 \\ \lambda_2(2x + y - 15) &= 0 \\ 2x - 4y - 6 + \lambda_1 + 2\lambda_2 &= 0 \\ -4x + 8y - 3 + \lambda_1 + \lambda_2 &= 0.\end{aligned}$$

If $\lambda_2 > 0$ and $\lambda_1 = 0$, then $2x + y = 15$. We add $2 \times$ the second-to-last equation to the last equation and get $\lambda_2 = 3$, which gives $2x - 4y = 0$. Hence we obtain $x = 6$ and $y = 3$ as an optimal solution to (3) by Theorem 9.30(ii).

(f) Is there more than one solution to (3)?

ANSWER:

We show that there cannot be solutions to the KKT conditions other than the one we found in the previous part. This shows by Theorem 9.30(i) that there can only be the one solution we found.

If $\lambda_1 = \lambda_2 = 0$, we get

$$\begin{aligned}2x - 4y &= 6 \\ -4x + 8y &= 3\end{aligned}$$

which cannot be solved. We also cannot have $\lambda_1 > 0$ and $\lambda_2 > 0$, since the system of equations

$$\begin{aligned}x + y &= 10 \\ 2x + y &= 15\end{aligned}$$

has the solution $x = y = 5$, which substituted into the last equation gives $17 + \lambda_1 + \lambda_2 = 0$, which contradicts $\lambda_1, \lambda_2 \geq 0$.

If $\lambda_1 > 0$, then $\lambda_2 = 0$. By multiplying the second-to-last equation by 2 and adding to the last equation, we see that $\lambda_1 = 5$. Hence

$$\begin{aligned}2x - 4y &= 1 \\ -4x + 8y &= -2\end{aligned}$$

i.e., $x + y = 10$ and $x = 2y + \frac{1}{2}$, which gives $y = \frac{19}{6}$ and $x = \frac{41}{6}$. But this contradicts $2x + y \leq 15$.

(g) Use a previous part to find a vector $v \in \mathbb{R}^2 \setminus \{0\}$ such that $f(tv) = tf(v)$ for $t \in \mathbb{R}$. Is f strictly convex?

ANSWER:

Inspired by the first part, we consider the vector $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Along the line tv , we have

$$f(2t, t) = f(tv) = -15t = tf(v).$$

That is,

$$f((1-t) \cdot 0 + tv) = -15t = (1-t)f(0) + tf(v)$$

and the function cannot be strictly convex according to Definition 6.1.

PROBLEM 3

Consider the system of linear equations

$$\begin{aligned}x + y + 2z &= 1 \\x + 2y + 3z &= 1 \\y + 2z &= t\end{aligned}\tag{4}$$

where $t \in \mathbb{R}$.

(a) The system of equations (4) can be written in matrix form $Av = b$, where

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 1 \\ 1 \\ t \end{pmatrix}.$$

Find the matrix A .

ANSWER:

The matrix is read from (4) to be

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}.$$

from the rules of matrix multiplication.

(b) Show that (4) can be solved uniquely for all $t \in \mathbb{R}$.

ANSWER:

By subtracting the first equation from the second equation, we get $y + z = 0$, which is then subtracted from the third equation with the result $z = t$. Substitution into $y + z = 0$ gives $y + t = 0 \implies y = -t$. Substitution into the first equation gives $x - t + 2t = 1 \implies x = 1 - t$. That is,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 - t \\ -t \\ t \end{pmatrix}$$

is the unique solution. It is also possible to use the next part to answer this part, since $v = A^{-1}b$.

(c) Show that

$$A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix},$$

where A is the matrix from (a).

ANSWER:

Here one can either verify that $AA^{-1} = A^{-1}A = I_3$ or compute A^{-1} from the method in §3.5.1 by transforming the matrix

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{pmatrix}.$$

(d) Let $f : \mathbb{R}^4 \rightarrow \mathbb{R}$ be given by

$$f(x, y, z, t) = (1 - x - y - 2z)^2 + (1 - x - 2y - 3z)^2 + (t - y - 2z)^2.$$

Find the global minima of f . Is f strictly convex?

ANSWER:

Here $f(x, y, z, t) \geq 0$ for all x, y, z, t and $f(x, y, z, t) = 0$ if and only if x, y, z, t satisfy (4). For each t , there is exactly one solution to (4), and hence infinitely many global minima for f . Therefore f cannot be strictly convex by Theorem 6.12.